

Exercise 7: The Qualifying Project

Congratulations, you made it to the end of our introductory training sequence! You've got all the tools you need to tackle a computer programming challenge. To demonstrate that you are ready to move on to a research project, I'd like you all to try to create some scientific modeling software. The goal is to build a dynamic simulation of a particle moving through 3D space under the influence of gravity, and then add some random noise to its path. You can then characterize the effects that various magnitudes and types of noise have on particle motion. Once you have completed this task, put together a short report (in word document, with figures) that summarizes your findings. Send me the report and your code, and we can meet individually to discuss it. Once you have completed this task, you'll be ready to join on of our research teams and work towards publishing a journal article.

This will be a challenging exercise. I expect it will take multiple research sessions to complete, and that most people will get stuck somewhere along the way. This is much more similar to real research than what you have been doing in previous training exercises. Welcome to the deep end of the pool! Good luck, and track me down if you get stuck.

Some things to consider:

- Work in 3D space. xyz coordinates are probably best because the system is not rotating. If you want to work in spherical polar coordinates for added flair, more power to you.
- You will need to set key parameters such as initial position, velocity, and acceleration, mass, etc. somewhere at the beginning of your program. These can be user-supplied or hard coded.
- Set the "ground level" somewhere in your program, and make sure your particle can't fall through the ground! We want our simulations realistic.
- Of course you can use Newton's equations of motion to calculate the position and velocity of a particle under constant acceleration at any point in time. You may want to write this program first, to get the basic physics straight. Equations are at the end of this list. You can't add random noise to this model in a physically realistic way though, and acceleration has to stay constant.

- To get good random noise, you will to make your particle take small steps in time. For each time step, you can calculate the expected change in its position, velocity, and acceleration parameters (decide which should be changing). Then you can propagate your particle's parameters in time by multiplying the time step by the derivative to find the change, then adding it to the previous value of the parameter. This is called the Euler Method, if you want to google it.
- You can find the random number library functions in chapter 26.7 of the online manual (<https://www.gnu.org/software/octave/doc/interpreter/Random-Number-Generation.html>).
- Some things you could add noise to: particle position in a given direction (modeling Brownian motion), force undergone by the particle in a given direction (inconsistent wind), mass of the particle (weird but interesting), the direction of a small force (air turbulence), the force of gravity (another weird one).
- I've left a couple of physics textbooks on the bench if you need reference material for particle trajectories. The basic equations for motion under constant acceleration are:

$$F = ma$$

$$x(t) = x_0 + v_0 t + \frac{1}{2}at^2$$

$$v(t) = v_0 + at$$